FAIRClinical: Making published clinical case reports and supplementary data FAIR

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What is a case report and what is the problem?

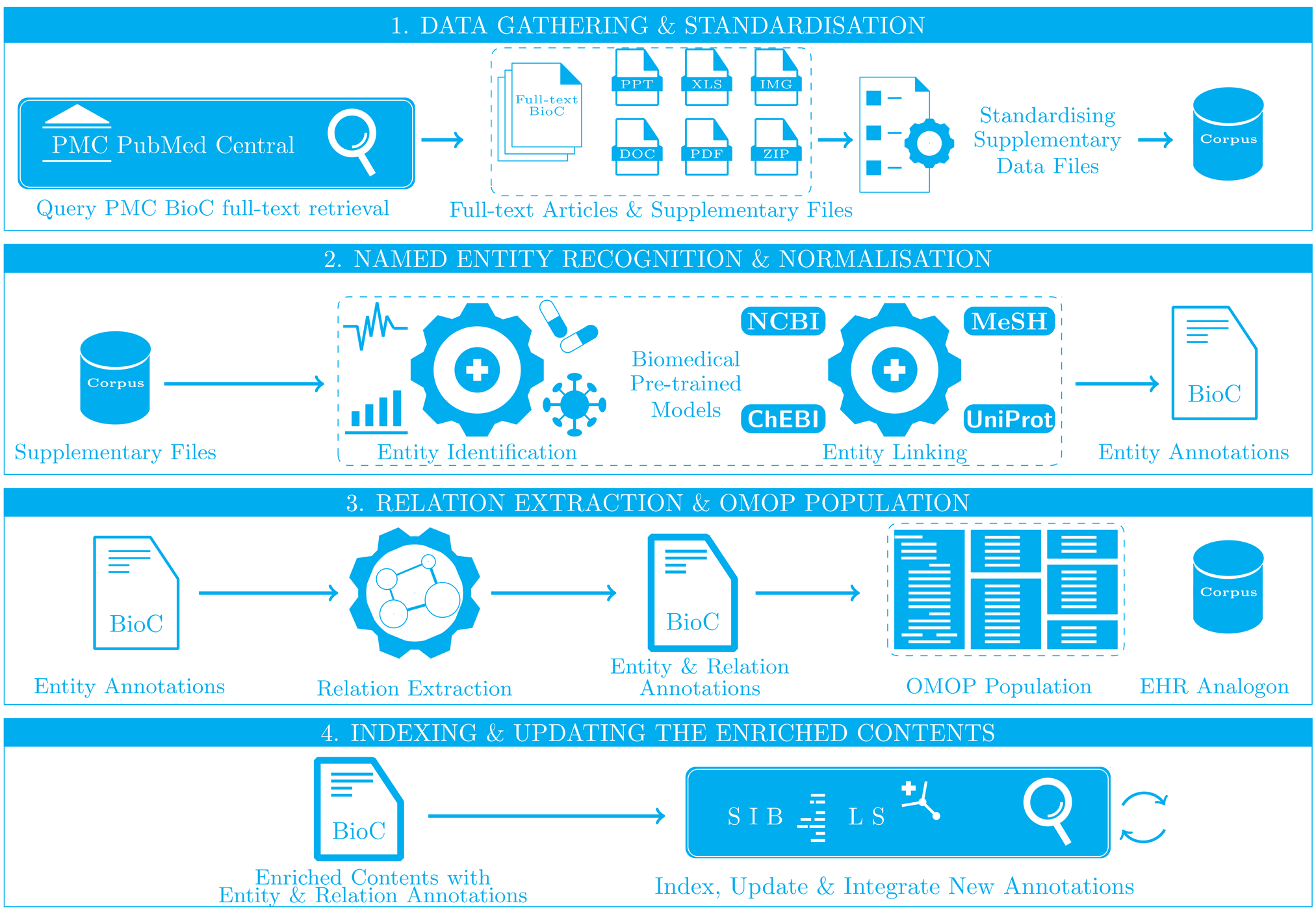
A clinical case report describes the medical signs and symptoms, diagnosis, treatment, and follow-up of an individual patient. These are published in the biomedical literature along with accompanying supplementary files in various formats (e.g., PDF, spreadsheet, word processor, text files). It is difficult for researchers to reuse this valuable collection of clinical data due to constraints searching unstructured text and heterogeneous supplementary file formats.

What is FAIRClinical doing about it?

The FAIRClinical project is enhancing the FAIR-ness (Findability, Accessibility, Interoperability, Reusability) of clinical case reports and supplementary data. State-of-the-art natural language processing (NLP) identifies medical concepts, normalises the entities using relevant terminologies, extracts relations among the entities, and standardises them into a Common Data Model (CDM).

What FAIR data resources are being produced?

An OMOP CDM reference database, populated with the standardised concepts from case reports, will be made openly available for the community to use as an Electronic Health Record (EHR)-like dataset. Supplementary files are annotated with three types of metadata: administrative (e.g., data licensing model), descriptive demographics (e.g., gender and age of patient), and structural (e.g., file format). These enriched supplementary data will be searchable from the SIB Literature Services (SIBiLS) via new infrastructure to openly share literature and supplementary data annotations.



Can this FAIRification approach be used for other scientific fields?

The FAIRClinical methodologies for putting FAIR data resources at the fingertips of researchers to find, compare, and reuse clinical data for research can be adapted for other life sciences fields. The transformer-based NLP models can be trained for other domains and used to populate community agreed data models. The SIBiLS supplementary metadata search is agnostic to the scientific field, so able to share supplementary data annotations across life sciences literature.

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